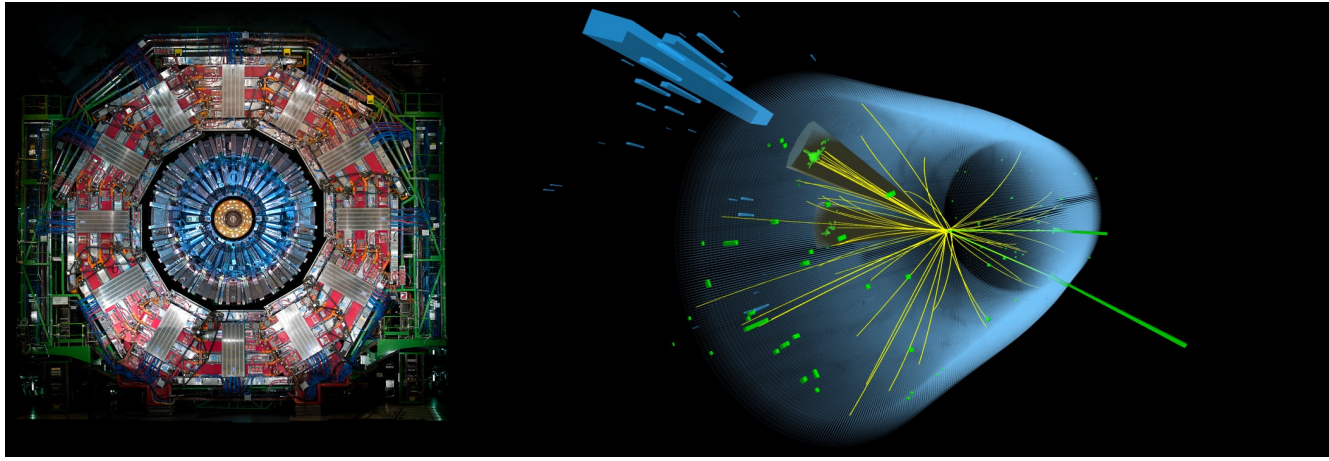




Particle Physics

Weak Interaction

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Parity

- The parity operator performs spatial inversion through the origin

$$\psi'(\vec{x}, t) = \hat{P}\psi(\vec{x}, t) = \psi(-\vec{x}, t)$$

Applying $\hat{P}$ twice : $\hat{P}\hat{P} = I \rightarrow \hat{P}^{-1} = \hat{P} \quad \dots (1)$

- If physics is invariant under parity transformation, $\langle \psi | \psi \rangle = \langle \psi' | \psi' \rangle$

$$\hat{P}^\dagger \hat{P} = I \quad \dots (2)$$

- So the parity operator is unitary and from (1) and (2), $\hat{P}^\dagger = \hat{P}$, therefore $\hat{P}$ is Hermitian operator
- If the interaction Hamiltonian commutes with $\hat{P}$, parity is an observable conserved quantity
- If $\psi(\vec{x}, t)$ is an eigenfunction of the parity operator with eigenvalue P

$$\hat{P}\psi(\vec{x}, t) = P\psi(\vec{x}, t) \rightarrow \hat{P}\hat{P}\psi(\vec{x}, t) = P\hat{P}\psi(\vec{x}, t) = P^2\psi(\vec{x}, t)$$

$$P^2 = 1$$

 Parity has eigenvalues $P = \pm 1$

- QED and QCD are invariant under parity
- Experimentally observe that Weak Interactions do not conserve parity

Intrinsic parities of fundamental particles

- **Spin-1 Bosons**
 - $P(\gamma) = P(g) = P(W^\pm) = P(Z) = -1$
- **Spin-1/2 Fermions**
 - $P(e^-) = P(\nu_e) = P(q) = +1$
 - $P(e^+) = P(\bar{\nu}_e) = P(\bar{q}) = -1$
 - **For Dirac spinors, the parity operator is:**

$$\hat{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Parity Conservation in QED and QCD

- Consider the QED process $e^- q \rightarrow e^- q$

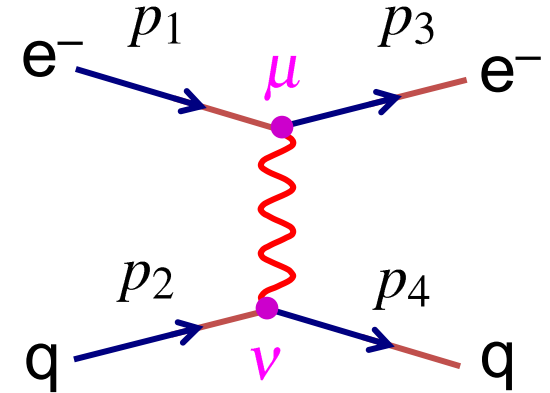
- The Feynman rules for QED gives:

$$-iM = [\bar{u}_e(p_3)ie\gamma^\mu u_e(p_1)] \frac{-ig_{\mu\nu}}{q^2} [\bar{u}_q(p_4)ie\gamma^\nu u_q(p_2)]$$

- Which can be expressed in terms of the electron and quark 4-vector currents:

$$M = -\frac{e^2}{q^2} g_{\mu\nu} j_e^\mu j_q^\nu = -\frac{e^2}{q^2} j_e \cdot j_q$$

With $j_e = \bar{u}_e(p_3)\gamma^\mu u_e(p_1)$ and $j_q = \bar{u}_q(p_4)\gamma^\mu u_q(p_2)$



- ✓ Consider what happens to the matrix element under the parity transformation

- Spinors transform as $u \xrightarrow{\hat{P}} \hat{P}u = \gamma^0 u$

- Adjoint spinors transform as

$$\bar{u} = u^\dagger \gamma^0 \xrightarrow{\hat{P}} (\hat{P}u)^\dagger \gamma^0 = u^\dagger \gamma^{0\dagger} \gamma^0 = u^\dagger \gamma^0 \gamma^0 = \bar{u} \gamma^0$$

$$\bar{u} \xrightarrow{\hat{P}} \bar{u} \gamma^0$$

- Hence $j_e = \bar{u}_e(p_3)\gamma^\mu u_e(p_1) \xrightarrow{\hat{P}} \bar{u}_e(p_3)\gamma^0 \gamma^\mu \gamma^0 u_e(p_1)$

- Consider the components of the four-vector current

0: $j_e^0 \xrightarrow{\hat{P}} \bar{u} \gamma^0 \gamma^0 \gamma^0 u = \bar{u} \gamma^0 u = j_e^0$ since $\gamma^0 \gamma^0 = 1$

$k=1,2,3$: $j_e^k \xrightarrow{\hat{P}} \bar{u} \gamma^0 \gamma^k \gamma^0 u = -\bar{u} \gamma^k \gamma^0 \gamma^0 u = -\bar{u} \gamma^k u = -j_e^k$ since $\gamma^0 \gamma^k = -\gamma^k \gamma^0$

- The time-like component remains unchanged and the space-like components change sign

Similarly $j_q^0 \xrightarrow{\hat{P}} j_q^0 \quad j_q^k \xrightarrow{\hat{P}} -j_q^k \quad k = 1, 2, 3$

- ✓ Consequently the four-vector scalar product

$$j_e \cdot j_q = j_e^0 j_q^0 - j_e^k j_q^k \xrightarrow{\hat{P}} j_e^0 j_q^0 - (-j_e^k)(-j_q^k) = j_e \cdot j_q$$

QED Matrix Elements are Parity Invariant



Parity Conserved in QED

- ✓ The QCD vertex has the same form and, thus,



Parity Conserved in QCD

or $j^\mu \xrightarrow{\hat{P}} j_\mu$

$$j^\mu \cdot j^\nu \xrightarrow{\hat{P}} j_\mu \cdot j_\nu$$

$$\xrightarrow{\hat{P}} j^\mu \cdot j^\nu$$

Parity Violation in β -Decay

- The parity operator $\hat{P}$ corresponds to a discrete transformation $x \rightarrow -x$, etc

Vectors
change sign

$$\left\{ \begin{array}{l} \vec{r} \xrightarrow{\hat{P}} -\vec{r} \\ \vec{p} \xrightarrow{\hat{P}} -\vec{p} \end{array} \right. \quad (p_x = \frac{\partial}{\partial x}, \text{ etc.})$$

Axial-Vectors
unchanged

$$\left\{ \begin{array}{l} \vec{L} \xrightarrow{\hat{P}} \vec{L} \\ \vec{\mu} \xrightarrow{\hat{P}} \vec{\mu} \end{array} \right. \quad \begin{array}{l} (\vec{L} = \vec{r} \times \vec{p}) \\ (\vec{\mu} \propto \vec{L}) \end{array}$$

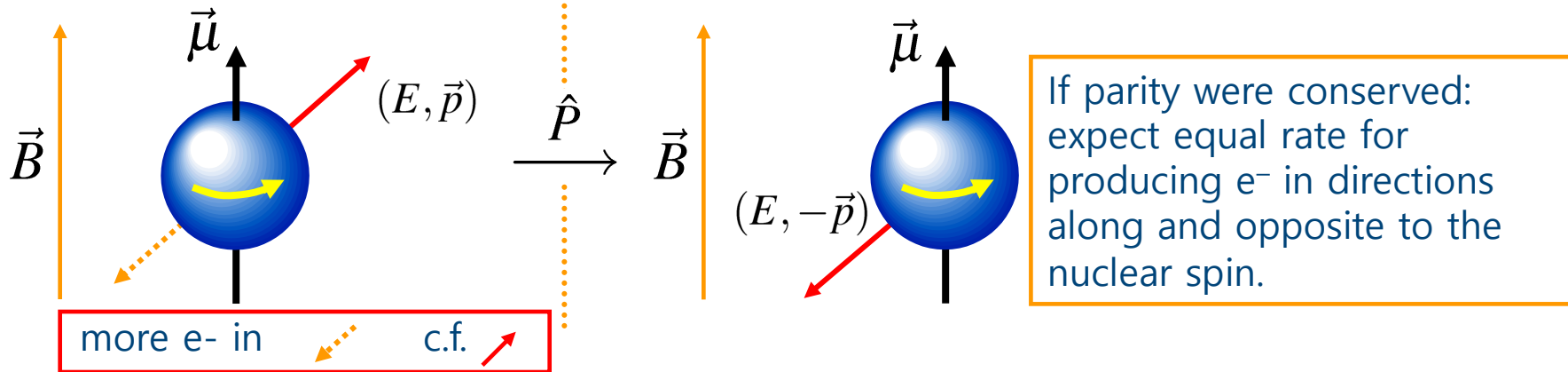
Note B is an axial vector

$$d\vec{B} \propto \vec{J} \times \vec{r} d^3\vec{r}$$

- 1957: C.S.Wu et al. studied beta decay of polarized cobalt-60 nuclei:



- Observed **electrons emitted preferentially** in direction opposite to applied field



- Conclude **parity is violated** in **WEAK INTERACTION**
 $\rightarrow$ that the **WEAK interaction** vertex is **NOT** of the form $\bar{u}_e \gamma^\mu u_\nu$

Bilinear Covariant

- ★ The requirement of Lorentz invariance of the matrix element severely restricts the form of the interaction vertex. QED and QCD are “**VECTOR**” interactions:

$$j^\mu = \bar{\psi} \gamma^\mu \phi$$

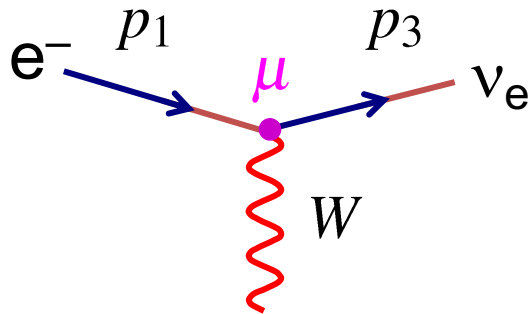
- ★ This combination transforms as a 4-vector
- ★ In general, there are only 5 possible combinations of two spinors and the gamma matrices that form Lorentz invariant currents, called “bilinear covariants”:

Type	Form	Components	“Boson Spin”
♦ SCALAR	$\bar{\psi} \phi$	1	0
♦ PSEUDOSCALAR	$\bar{\psi} \gamma^5 \phi$	1	0
♦ VECTOR	$\bar{\psi} \gamma^\mu \phi$	4	1
♦ AXIAL VECTOR	$\bar{\psi} \gamma^\mu \gamma^5 \phi$	4	1
♦ TENSOR	$\bar{\psi} (\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) \phi$	6	2

- ★ Note that in total the sixteen components correspond to the 16 elements of a general 4x4 matrix: “decomposition into Lorentz invariant combinations”
- ★ In QED the factor $g_{\mu\nu}$ arose from the sum over polarization states of the virtual photon (2 transverse + 1 longitudinal, 1 scalar) = (2J+1) + 1
- ★ Associate SCALAR and PSEUDOSCALAR interactions with the exchange of a SPIN-0 boson, etc. – no spin degrees of freedom

V-A Structure of the Weak Interaction

- The most general form for the interaction between a fermion and a boson is a linear combination of bilinear covariants
- For an interaction corresponding to the exchange of a spin-1 particle, the most general form is a linear combination of VECTOR and AXIAL-VECTOR
- The form for WEAK interaction is determined from experiment to be VECTOR-AXIAL-VECTOR (V-A)



$$j^\mu \propto \bar{u}_{\nu_e} (\gamma^\mu - \gamma^\mu \gamma^5) u_e$$

V - A

- Can this account for parity violation?
- First consider parity transformation of a pure AXIAL-VECTOR current

$$j_A = \bar{\psi} \gamma^\mu \gamma^5 \phi \quad \text{with} \quad \gamma^5 = i\gamma^0 \gamma^1 \gamma^2 \gamma^3; \quad \gamma^5 \gamma^0 = -\gamma^0 \gamma^5$$

$$j_A = \bar{\psi} \gamma^\mu \gamma^5 \phi \xrightarrow{\hat{P}} \bar{\psi} \gamma^0 \gamma^\mu \gamma^5 \gamma^0 \phi = -\bar{\psi} \gamma^0 \gamma^\mu \gamma^0 \gamma^5 \phi$$

$$j_A^0 \xrightarrow{\hat{P}} -\bar{\psi} \gamma^0 \gamma^0 \gamma^0 \gamma^5 \phi = -\bar{\psi} \gamma^0 \gamma^5 \phi = -j_A^0$$

$$j_A^k \xrightarrow{\hat{P}} -\bar{\psi} \gamma^0 \gamma^k \gamma^0 \gamma^5 \phi = +\bar{\psi} \gamma^k \gamma^5 \phi = +j_A^k \quad k = 1, 2, 3$$

$$\text{or} \quad j_A^\mu \xrightarrow{\hat{P}} -j_{A\mu}$$

- The space-like components remain unchanged and the time-like components change sign (the opposite to the parity properties of a vector-current)

$$\boxed{j_A^0 \xrightarrow{\hat{P}} -j_A^0; \quad j_A^k \xrightarrow{\hat{P}} +j_A^k; \quad j_V^0 \xrightarrow{\hat{P}} +j_V^0; \quad j_V^k \xrightarrow{\hat{P}} -j_V^k}$$

- Now consider the matrix elements

$$M \propto g_{\mu\nu} j_1^\mu j_2^\nu = j_1^0 j_2^0 - \sum_{k=1,3} j_1^k j_2^k$$

- For the combination of a two axial-vector currents

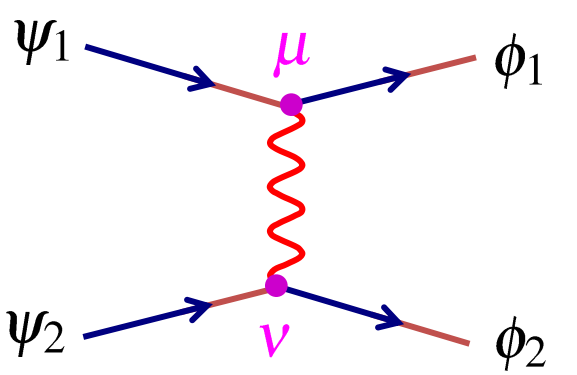
$$j_{A1} \cdot j_{A2} \xrightarrow{\hat{P}} (-j_1^0)(-j_2^0) - \sum_{k=1,3} (j_1^k)(j_2^k) = j_{A1} \cdot j_{A2}$$

- Consequently parity is conserved for both a pure vector and pure axial-vector interactions
- However the combination of a vector current and an axial vector current

$$j_{V1} \cdot j_{A2} \xrightarrow{\hat{P}} (j_1^0)(-j_2^0) - \sum_{k=1,3} (-j_1^k)(j_2^k) = -j_{V1} \cdot j_{A2}$$

changes sign under parity – can give parity violation !

- ★ Now consider a general linear combination of VECTOR and AXIAL-VECTOR
(note this is relevant for the Z-boson vertex)



$$j_1 = \bar{\phi}_1 (g_V \gamma^\mu + g_A \gamma^\mu \gamma^5) \psi_1 = g_V j_1^V + g_A j_1^A$$

$$\frac{g_{\mu\nu}}{q^2 - m^2}$$

$$j_2 = \bar{\phi}_2 (g_V \gamma^\mu + g_A \gamma^\mu \gamma^5) \psi_2 = g_V j_2^V + g_A j_2^A$$

$$M_{fi} \propto j_1 \cdot j_2 = g_V^2 j_1^V \cdot j_2^V + g_A^2 j_1^A \cdot j_2^A + g_V g_A (j_1^V \cdot j_2^A + j_1^A \cdot j_2^V)$$

- Consider the parity transformation of this scalar product

$$j_1 \cdot j_2 \xrightarrow{\hat{P}} g_V^2 j_1^V \cdot j_2^V + g_A^2 j_1^A \cdot j_2^A - g_V g_A (j_1^V \cdot j_2^A + j_1^A \cdot j_2^V)$$

- If either g_A or g_V is zero, Parity is conserved, i.e. parity conserved in a pure VECTOR or pure AXIAL-VECTOR interaction

- Relative strength of parity violating part $\propto \frac{g_V g_A}{g_V^2 + g_A^2}$

Maximal Parity Violation for V-A (or V+A)

Chiral Structure of QED (Reminder)

- ★ Recall : introduced CHIRAL projections operators

$$P_R = \frac{1}{2}(1 + \gamma^5); \quad P_L = \frac{1}{2}(1 - \gamma^5)$$

project out chiral right- and left- handed states

- ★ In the ultra-relativistic limit, chiral states correspond to helicity states

- ★ Any spinor can be expressed as:

$$\psi = \frac{1}{2}(1 + \gamma^5)\psi + \frac{1}{2}(1 - \gamma^5)\psi = P_R\psi + P_L\psi = \psi_R + \psi_L$$

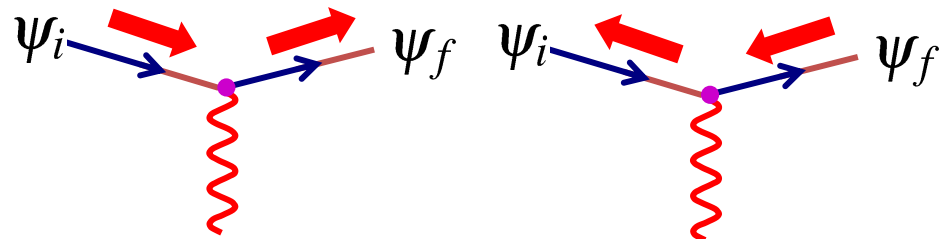
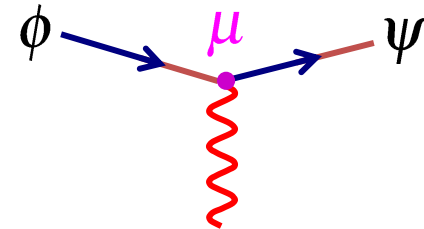
- The QED vertex $\bar{\psi}\gamma^\mu\phi$ in terms of chiral states:

$$\bar{\psi}\gamma^\mu\phi = \bar{\psi}_R\gamma^\mu\phi_R + \bar{\psi}_R\gamma^\mu\phi_L + \bar{\psi}_L\gamma^\mu\phi_R + \bar{\psi}_L\gamma^\mu\phi_L$$

conserves chirality, e.g.

$$\begin{aligned} \bar{\psi}_R\gamma^\mu\phi_L &= \frac{1}{2}\psi^\dagger(1 + \gamma^5)\gamma^0\gamma^\mu\frac{1}{2}(1 - \gamma^5)\phi \\ &= \frac{1}{4}\psi^\dagger\gamma^0(1 - \gamma^5)\gamma^\mu(1 - \gamma^5)\phi \\ &= \frac{1}{4}\bar{\psi}\gamma^\mu(1 + \gamma^5)(1 - \gamma^5)\phi = 0 \end{aligned}$$

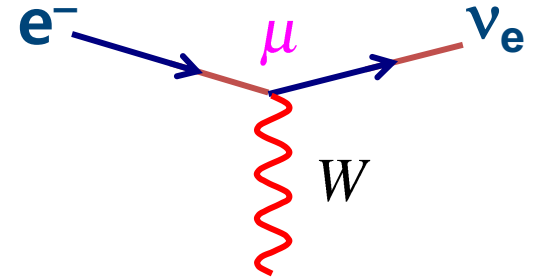
- ★ In the ultra-relativistic limit only two helicity combinations are non-zero



Helicity Structure of the WEAK Interaction

★ The charged current ($W^\pm$) weak vertex is:

$$\frac{-ig_w}{\sqrt{2}} \frac{1}{2} \gamma^\mu (1 - \gamma^5)$$

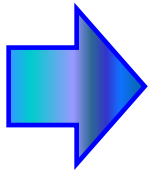


★ Since $\frac{1}{2}(1 - \gamma^5)$ projects out left-handed chiral particle states:

$$\bar{\psi} \frac{1}{2} \gamma^\mu (1 - \gamma^5) \phi = \bar{\psi} \gamma^\mu \phi_L$$

★ Writing $\bar{\psi} = \bar{\psi}_R + \bar{\psi}_L$ and from discussion of QED, $\bar{\psi}_R \gamma^\mu \phi_L = 0$ gives

$$\bar{\psi} \frac{1}{2} \gamma^\mu (1 - \gamma^5) \phi = \bar{\psi}_L \gamma^\mu \phi_L$$



Only the left-handed chiral components of particle spinors and right-handed chiral components of anti-particle spinors participate in charged current weak interactions

★ At very high energy ($E \gg m$), the left-handed chiral components are helicity eigenstates:

$$\frac{1}{2}(1 - \gamma^5)u \Rightarrow$$

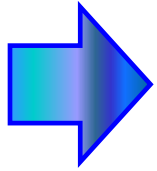


LEFT-HANDED PARTICLES
Helicity = -1

$$\frac{1}{2}(1 - \gamma^5)v \Rightarrow$$

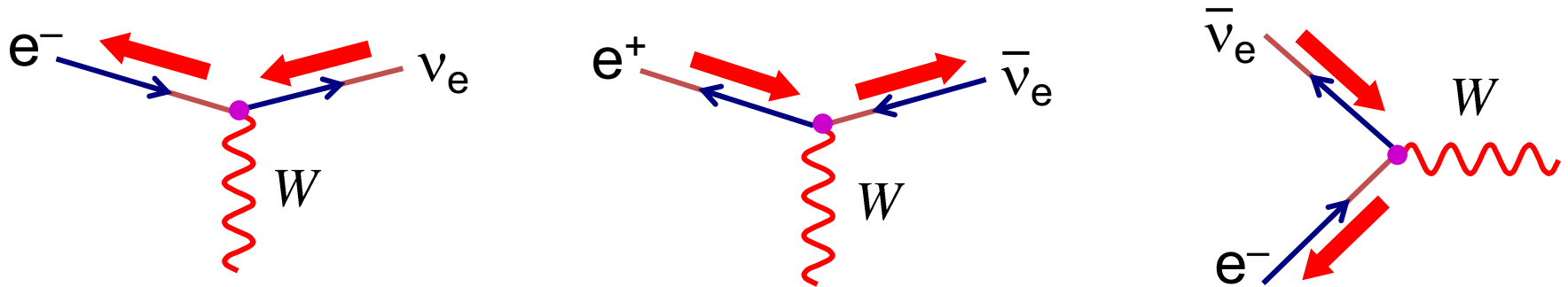


RIGHT-HANDED ANTI-PARTICLES
Helicity = +1



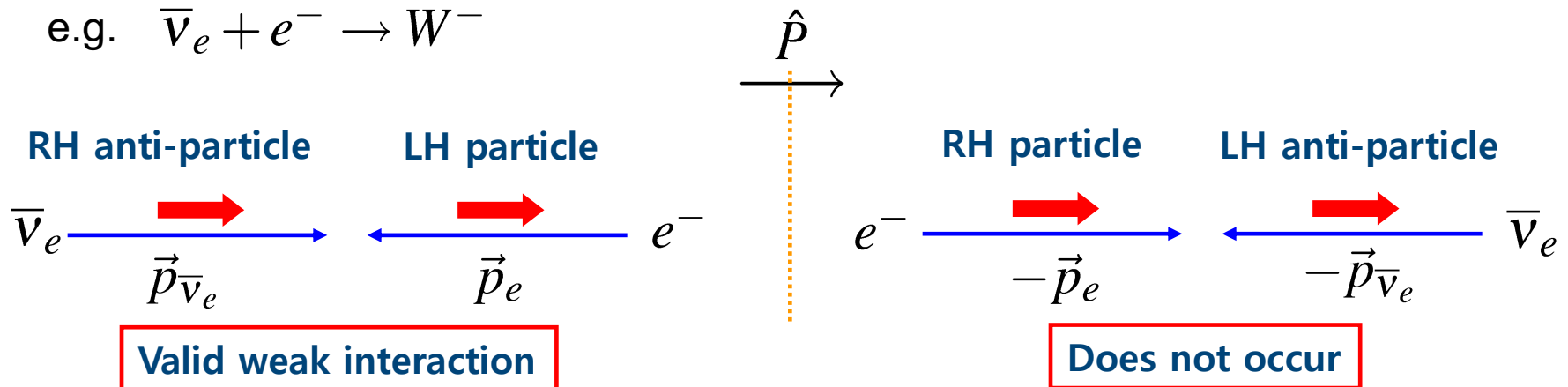
In the ultra-relativistic limit only **left-handed particles** and **right-handed antiparticles** participate in charged current weak interactions

e.g. In the relativistic limit, the only possible electron – neutrino interactions are:



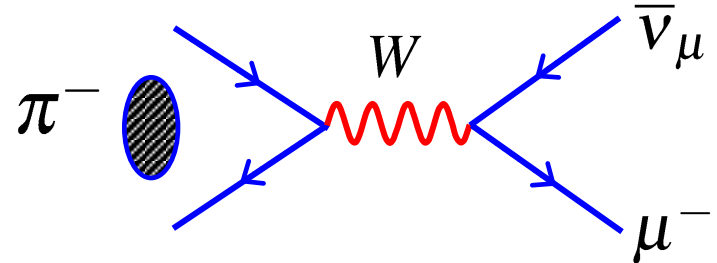
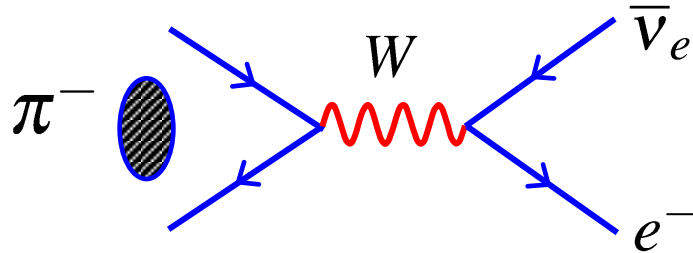
★ The helicity dependence of the weak interaction $\longleftrightarrow$ parity violation

e.g. $\bar{\nu}_e + e^- \rightarrow W^-$



Helicity in Pion Decay

- ★ The decays of charged pions provide a good demonstration of the role of helicity in the weak interaction



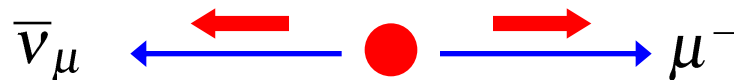
EXPERIMENTALLY:

$$\frac{\Gamma(\pi^- \rightarrow e^- \bar{\nu}_e)}{\Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu)} = 1.23 \times 10^{-4}$$

- Might expect the decay to electrons to dominate – due to increased phase space.... The opposite happens, the electron decay is helicity suppressed

- ★ Consider decay in pion rest frame.

- Pion is spin zero: so the spins of the ν and muon are opposite
- Weak interaction only couples to RH chiral anti-particle states. Since neutrinos are (almost) massless, must be in RH Helicity state
- Therefore, to conserve angular mom. muon is emitted in a RH HELICITY state



- But only left-handed CHIRAL particle states participate in weak interaction

★The general **right-handed helicity** solution to the Dirac equation is

$$u_{\uparrow} = N \begin{pmatrix} c \\ e^{i\phi} s \\ \frac{|\vec{p}|}{E+m} c \\ \frac{|\vec{p}|}{E+m} e^{i\phi} s \end{pmatrix} \quad \text{with} \quad c = \cos \frac{\theta}{2} \quad \text{and} \quad s = \sin \frac{\theta}{2}$$

- project out the **left-handed chiral** part of the wave-function using

$$P_L = \frac{1}{2}(1 - \gamma^5) = \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

giving

$$P_L u_{\uparrow} = \frac{1}{2} N \left(1 - \frac{|\vec{p}|}{E+m} \right) \begin{pmatrix} c \\ e^{i\phi} s \\ -c \\ -e^{i\phi} s \end{pmatrix} = \frac{1}{2} N \left(1 - \frac{|\vec{p}|}{E+m} \right) u_L$$

In the limit $m \ll E$ this tends to zero

- similarly

$$P_R u_{\uparrow} = \frac{1}{2} N \left(1 + \frac{|\vec{p}|}{E+m} \right) \begin{pmatrix} c \\ e^{i\phi} s \\ c \\ e^{i\phi} s \end{pmatrix} = \frac{1}{2} N \left(1 + \frac{|\vec{p}|}{E+m} \right) u_R$$

In the limit $m \ll E$, $P_R u_{\uparrow} \rightarrow u_R$

★Hence

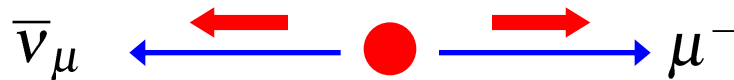
$$\boxed{u_{\uparrow}} = P_R u_{\uparrow} + P_L u_{\uparrow} = \frac{1}{2} \left(1 + \frac{|\vec{p}|}{E+m} \right) \boxed{u_R} + \frac{1}{2} \left(1 - \frac{|\vec{p}|}{E+m} \right) \boxed{u_L}$$

RH Helicity

RH Chiral

LH Chiral

- In the limit $E \gg m$, as expected, the RH chiral and helicity states are identical
- Although only LH chiral particles participate in the weak interaction the contribution from RH Helicity states is not necessarily zero !



$m_n \approx 0$: RH Helicity $\equiv$ RH Chiral

$m_m \neq 0$: RH Helicity has LH Chiral Component

- ★ Expect matrix element to be proportional to LH chiral component of RH Helicity electron/muon spinor

$$M_{fi} \propto \frac{1}{2} \left(1 - \frac{|\vec{p}|}{E+m} \right) = \boxed{\frac{m_{\mu}}{m_{\pi} + m_{\mu}}}$$

from the kinematics of pion decay at rest

- ★ Hence because the electron mass is much smaller than the pion mass the decay $\pi^- \rightarrow e^- \bar{\nu}_e$ is heavily suppressed.

Evidence for V-A

★The V-A nature of the charged current weak interaction vertex fits with experiment

EXAMPLE charged pion decay

•Experimentally measure:

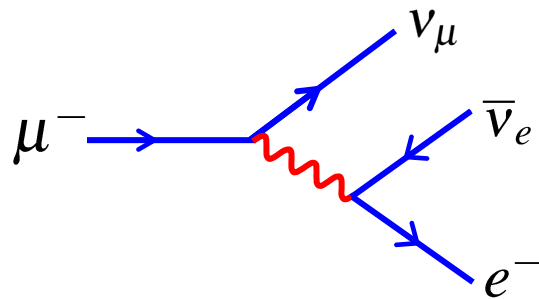
$$\frac{\Gamma(\pi^- \rightarrow e^- \bar{\nu}_e)}{\Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu)} = (1.230 \pm 0.004) \times 10^{-4}$$

•Theoretical predictions (depend on Lorentz Structure of the interaction)

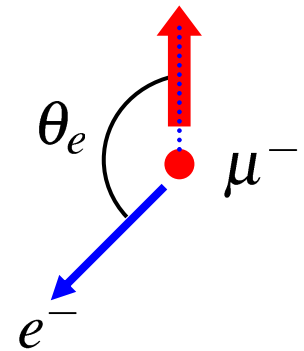
V-A $(\bar{\psi}\gamma^\mu(1-\gamma^5)\phi)$ or **V+A** $(\bar{\psi}\gamma^\mu(1+\gamma^5)\phi)$ $\frac{\Gamma(\pi^- \rightarrow e^- \bar{\nu}_e)}{\Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu)} \approx 1.3 \times 10^{-4}$

Scalar $(\bar{\psi}\phi)$ or **Pseudo-Scalar** $(\bar{\psi}\gamma^5\phi)$ $\frac{\Gamma(\pi^- \rightarrow e^- \bar{\nu}_e)}{\Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu)} = 5.5$

EXAMPLE muon decay



Measure **electron** energy and angular distributions relative to muon spin direction. Results expressed in terms of general **S+P+V+A+T** form in "Michel Parameters"



e.g. TWIST expt: 6×10^9 m decays
Phys. Rev. Lett. 95 (2005) 101805

V-A Prediction:

$$\rho = 0.75080 \pm 0.00105$$

$$\rho = 0.75$$

Weak Charged Current Propagator

- ★ The charged-current Weak interaction is different from QED and QCD in that it is mediated by massive W-bosons (80.3 GeV)
- ★ This results in a more complicated form for the propagator:
 - in handout 4 showed that for the exchange of a massive particle:

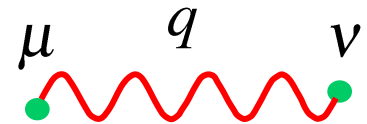
$$\begin{array}{ccc} \text{massless} & & \text{massive} \\ \frac{1}{q^2} & \longrightarrow & \frac{1}{q^2 - m^2} \end{array}$$

- In addition the sum over W boson polarization states modifies the numerator

● W-boson propagator

spin 1 $W^\pm$

$$\frac{-i \left[g_{\mu\nu} - q_\mu q_\nu / m_W^2 \right]}{q^2 - m_W^2}$$



- ★ However in the limit where q^2 is small compared with the interaction takes a simpler form.

$$m_W = 80.3 \text{ GeV}$$

● W-boson propagator ($q^2 \ll m_W^2$)

$$\frac{i g_{\mu\nu}}{m_W^2}$$



- The interaction appears point-like (i.e no q^2 dependence)

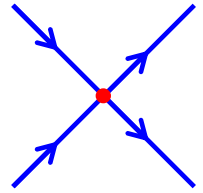
Connection to Fermi Theory

- ★ In 1934, before the discovery of parity violation, Fermi proposed, in analogy with QED, that the invariant matrix element for β -decay was of the form:

$$M_{fi} = G_F g_{\mu\nu} [\bar{\psi} \gamma^\mu \psi] [\bar{\psi} \gamma^\nu \psi]$$

where $G_F = 1.166 \times 10^{-5} \text{ GeV}^{-2}$

- Note the absence of a propagator : i.e. this represents an interaction at a point



- ★ After the discovery of parity violation in 1957 this was modified to

$$M_{fi} = \frac{G_F}{\sqrt{2}} g_{\mu\nu} [\bar{\psi} \gamma^\mu (1 - \gamma^5) \psi] [\bar{\psi} \gamma^\nu (1 - \gamma^5) \psi]$$

(the factor of $\sqrt{2}$ was included so the numerical value of G_F did not need to be changed)

- ★ Compare to the prediction for W-boson exchange

$$M_{fi} = [\frac{g_W}{\sqrt{2}} \bar{\psi} \frac{1}{2} \gamma^\mu (1 - \gamma^5) \psi] \frac{g_{\mu\nu} - q_\mu q_\nu / m_W^2}{q^2 - m_W^2} [\frac{g_W}{\sqrt{2}} \bar{\psi} \frac{1}{2} \gamma^\nu (1 - \gamma^5) \psi]$$

which for $q^2 \ll m_W^2$ becomes:

$$M_{fi} = \frac{g_W^2}{8m_W^2} g_{\mu\nu} [\bar{\psi} \gamma^\mu (1 - \gamma^5) \psi] [\bar{\psi} \gamma^\nu (1 - \gamma^5) \psi]$$

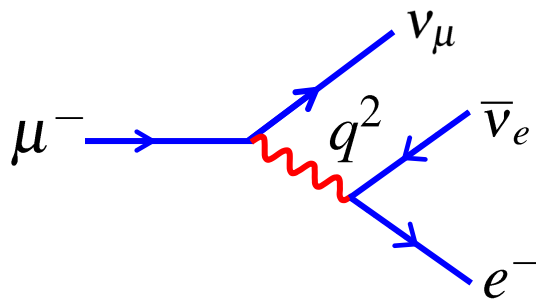


$$\frac{G_F}{\sqrt{2}} = \frac{g_W^2}{8m_W^2}$$

Still usually use G_F to express strength of weak interaction as this is the quantity that is precisely determined in muon decay

Strength of Weak Interaction

★ Strength of weak interaction most precisely measured in muon decay



• Here $q^2 < m_\mu$ (0.106 GeV)

• To a very good approximation the W-boson propagator can be written

$$\frac{-i [g_{\mu\nu} - q_\mu q_\nu / m_W^2]}{q^2 - m_W^2} \approx \frac{ig_{\mu\nu}}{m_W^2} \quad \boxed{\frac{G_F}{\sqrt{2}} = \frac{g_W^2}{8m_W^2}}$$

• In muon decay : $\Gamma(\mu^- \rightarrow e^- \nu_\mu \bar{\nu}_e) = \frac{1}{\tau_\mu} = \frac{G_F^2 m_\mu^5}{192\pi^3}$

• Muon lifetime and mass $\rightarrow G_F = 1.16639(1) \times 10^{-5} \text{ GeV}^{-2}$

★ To obtain the intrinsic strength of weak interaction need to know mass of W-boson: $m_W = 80.403 \pm 0.029 \text{ GeV}$

$$\rightarrow \alpha_W = \frac{g_W^2}{4\pi} = \frac{8m_W^2 G_F}{4\sqrt{2}\pi} = \frac{1}{30}$$



The intrinsic strength of the weak interaction is similar to, but greater than, the EM interaction ! It is the massive W-boson in the propagator which makes it appear weak. For $q^2 \gg m_W^2$ weak interactions are more likely than EM

Summary

- ★ Weak interaction is of form Vector – Axial-vector (V-A)

$$\frac{-ig_w}{\sqrt{2}} \frac{1}{2} \gamma^\mu (1 - \gamma^5)$$

- ★ Consequently only left-handed chiral particle states and right-handed chiral anti-particle states participate in the weak interaction



MAXIMAL PARITY VIOLATION

- ★ Weak interaction also violates Charge Conjugation symmetry
- ★ At low q^2 weak interaction is only weak because of the large W-boson mass

$$\frac{G_F}{\sqrt{2}} = \frac{g_W^2}{8m_W^2}$$

- ★ Intrinsic strength of weak interaction is similar to that of QED